Transition Grammar of the Voynich Manuscript: Sequential Constraints and Bidirectional Self-Clustering Symmetry

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Abstract

The Voynich Manuscript (Beinecke MS 408) is a 15th-century codex written in an undeciphered script. This paper reports a statistical analysis of sequential constraints, token-family structure, and bidirectional self-clustering in the Zandbergen–Landini EVA transliteration (31 608 tokens, 184 pages). Several findings are robust across sections, scribal hands, line-position controls, and frequency-conditioned tests. First, two distributed transition rules recur across the corpus: CHEDY→QOK attraction (2.625× above chance) and AIIN→QOK repulsion (0.504×), each spanning hundreds of unique token pairs and operating at the family level rather than as fixed collocations. Second, AIIN density is statistically invariant at 15.0% across Currier A and B pages (KS $p = 0.742$), a property no other family shares. Third, a bidirectional self-clustering test shows that Voynich is the only tested system with elevated, balanced clustering at both word boundaries (prefix 1.52×, suffix 1.54×, ratio 0.99). An earlier claim that Arabic is the closest structural match, based on prefix-only analysis, is retracted; under bidirectional analysis Arabic is strongly suffix-dominant. Ottoman Turkish was tested separately using a small historical corpus and also falls outside the Voynich profile (SYMM-LOW). Additional analyses identify line-bounded grammatical structure, productive morphology-like paradigms, and agreement-like feature propagation across adjacent and near-adjacent tokens. Cross-transcription validation across four independent transcribers (Currier, FSG, Takahashi, Grove) confirms that all core findings — bidirectional symmetry, transition rules, and line-bounded grammar — are stable regardless of tokenization decisions. These findings do not identify the manuscript's language, assign meaning to any token, or establish a decipherment. They do, however, define a set of structural constraints that any successful explanation must satisfy and make randomness, simple template generation, and simple substitution ciphers substantially less plausible. The strongest current explanation is that the manuscript encodes a structured language system, plausibly natural language, while imposing additional boundary structure not matched by the tested comparators.

Keywords: Voynich manuscript, computational linguistics, statistical analysis, morphological analysis, historical cryptography.

Reproducibility. All scripts, frozen datasets, and result files are available at https://github.com/amy2213/Voynich-Transition-Grammar. The canonical values for the three core findings (transition rules, AIIN invariance, bidirectional self-clustering) are verified by automated regression tests on every commit. Several additional analyses reported here are documented in the project's durable findings framework but are not yet packaged as standalone scripts; see §6.

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1 Introduction

The Voynich Manuscript is a 225-page codex of unknown authorship, written in an unknown script, that has resisted every attempt at decipherment since its emergence in the documentary record in the early seventeenth century. Vellum radiocarbon dating places its production between 1404 and 1438 (Hodgins, 2011). The physical manuscript is held at Yale's Beinecke Rare Book and Manuscript Library as MS 408 (Clemens, 2016). The text runs to approximately 38 000 word-like tokens across botanical, biological, cosmological, and pharmaceutical sections, each accompanied by illustrations. Despite detailed attention from professional cryptographers including William and Elizebeth Friedman, John Tiltman, and Prescott Currier, none of the many proposed decipherments has been independently verified.

Two broad research traditions have developed around the text. The first, pursued by cryptographers and linguists from the mid-twentieth century onward, attempts to identify the source language or cipher scheme directly. The second, more recent tradition treats the text as a data object and measures its statistical properties without attempting translation. Work in this second tradition has established that the text is non-random (Reddy and Knight, 2011), exhibits Zipfian type-token distribution (Bowern and Lindemann, 2021), contains semantically coherent keyword distributions (Montemurro and Zanette, 2013), and shows section-specific vocabulary with shared grammatical structure (Bowern and Lindemann, 2021). What remains unresolved is the generative mechanism: whether the token sequences reflect a natural language under some encoding, a constructed symbolic system, or a sophisticated cipher without a known 15th-century precedent.

This paper contributes to the second tradition. We analyze sequential constraints between token families at the class level — five families defined by EVA-transliteration affix patterns, plus a catch-all class — and compare the manuscript's structural profile against 16 natural-language comparators across 9 language families and a shuffled-token control. Five findings structure our contribution: (i) distributed transition rules operating at the family rather than token level; (ii) a bidirectional self-clustering test that removes the directional bias of prefix-only analyses; (iii) line-bounded grammatical structure that resets at every line boundary; (iv) agreement-like feature propagation across adjacent and near-adjacent tokens, including over intervening filler-like tokens; and (v) productive morphology-like paradigms with frequency-correlated variant generation. The bidirectional self-clustering framework reveals that Voynich occupies a region no other tested system does — symmetric-high — and that the previously reported Arabic match is an artifact of the prefix-only method.

We then synthesize the results into a *minimum viable explanation checklist*: eight quantitative properties that any proposed mechanism for producing the Voynich text must simultaneously satisfy. The checklist is a methodological device rather than a theoretical claim. It does not identify a single explanation as correct. It narrows the space of candidate explanations by making their requirements explicit and testable, and rules out classes of explanation (random generation, simple table schemes, monoalphabetic and polyalphabetic substitution ciphers) that cannot jointly satisfy the requirements.

We are careful about what this paper does not claim. It does not identify the manuscript's language. Structural similarity is not linguistic identification, and the comparator set, while substantial, does not cover every natural-language type — Sinitic, polysynthetic, and historical-shorthand systems are conspicuously absent and are listed as open problems. It does not decode any passage of the text or assign meaning to any token. And it does not argue that the manuscript is or is not a hoax; it argues that if it is a hoax, the mechanism must be substantially more sophisticated than proposed in earlier hoax theories.

§2 describes the data, the token family definitions, and the statistical methods. §3 reports findings under eleven subsections organized from the foundational (non-random structure, transition rules, AIIN invariance) through the cross-linguistic comparison to the more detailed analyses of line structure, suffix-agreement cascades, paradigm morphology, and glyph-layer architecture. §4 situates these findings against prior work. §5 presents the minimum viable explanation checklist. §6 and §7 list limitations and open problems.

2 Data and Methods

2.1 Core dataset

Analysis uses the Zandbergen–Landini transliteration of the Voynich Manuscript (Beinecke MS 408) in EVA (extended) alphabet, accessed via the `AncientLanguages/Voynich` dataset on Hugging Face and frozen as a local parquet snapshot. The corpus contains 4 197 lines, 31 608 tokens, and 184 pages. A canonical snapshot with SHA-256 checksum is included in the project repository; all analysis in this paper can be reproduced by a single command (`python run_all.py`) and is verified by automated regression tests.

Page attribution to Currier hands (A vs. B) and scribal hands (1–4) follows the metadata supplied in the source dataset. Pages are assigned to sections (Herbal A, Herbal B, Biological, Astronomical, Recipes Q20) using the folio numbering convention described in the canonical Voynich literature.

2.2 Token families

Tokens are classified into five non-overlapping families plus a catch-all `OTHER` class:

- **QOK**: tokens starting with `qok-`
- **OK**: tokens starting with `ok-` but not `qok-`
- **OT**: tokens starting with `ot-`
- **CHEDY**: tokens containing any of {`chedy`, `shedy`, `chey`, `shey`}
- **AIIN**: tokens containing `aiin` or `ain`

These definitions are EVA-specific and may not correspond to paleographic character boundaries in the original script. Results under FSG or Currier transcription alphabets are not tested here and remain an open question.

2.3 Comparison corpora

We compare Voynich against 16 natural-language comparators: 2 historical literary texts (Middle English Chaucer, Gutenberg #22120; KJV English, Gutenberg #10900) and 13 Wikipedia-derived corpora from the Leipzig Corpora Collection (Turkish, Hungarian, Finnish, Hebrew, Arabic, Latin, North Azerbaijani, Italian, Estonian, Swahili, Georgian, Tagalog, Mandarin Chinese; all `wikipedia_2021_100K` editions). A shuffled-token control (`Gibberish`) is derived from the Voynich corpus itself by random permutation. An additional small corpus (Ottoman Turkish, 16,890 words, UD treebank) is tested separately given its relevance as a potential historical match.

The Leipzig corpora are modern-language proxies, not medieval texts. This is an explicit limitation, discussed in §6.

2.4 Transition-rule analysis

For each class pair (s,d) we compute the observed/expected ratio

$$R(s \to d) = \frac{N_{s\to d}}{N_s \cdot (N_d/N_{\text{total}})},$$

where $N_{s\to d}$ is the count of adjacent (s,d) transitions, N_s and N_d are the marginal counts of s in source position and d in destination position, and N_{total} is the total number of transitions. Significance is assessed via permutation: the class sequence is shuffled 2,000 times and the observed ratio is compared against the shuffle distribution. For the full transition matrix, a χ^2 test of independence is computed on the 6×6 class transition table.

2.5 AIIN invariance test

Per-page AIIN density is computed for all pages with ≥ 20 tokens. Currier A pages ($n = 102$) and Currier B pages ($n = 72$) are compared via a two-sample Kolmogorov–Smirnov test. The mean difference is estimated with a 5,000-sample bootstrap.

2.6 Bidirectional self-clustering

For each system we compute two self-clustering scores:

- **Prefix SC:** mean $R(f \to f)$ over prefix-defined families. For Voynich, these are the EVA prefix families above. For comparison languages, the top-5 prefixes of 2–3 characters with coverage 2–20% are auto-detected.
- **Suffix SC:** the same computation using suffix-defined families, auto-detected analogously.

The prefix/suffix ratio P/S classifies each system into one of four buckets:

- **SYMM-HIGH**: both $> 1.1\times$, $0.80 \leq P/S \leq 1.25$
- **SUFFIX-DOM**: $S > 1.1\times$ and $P/S < 0.80$
- **PREFIX-DOM**: $P > 1.1\times$ and $P/S > 1.25$
- **SYMM-LOW**: both $< 1.1\times$

Bootstrap 95% confidence intervals are computed by resampling tokens with replacement (2,000 replicates).

2.7 Token-level grammar test

To distinguish a distributed class-level rule from a handful of fixed phrases, for each CHEDY token with ≥ 5 occurrences we compute the ratio of observed QOK-following to chance. Tokens with ratio > 1.3 are classified as "attractors." The proportion of attracting tokens, and the number of unique CHEDY→QOK token pairs, together diagnose whether the rule operates at the class level or on specific collocations.

2.8 Software and reproducibility

All analyses use Python 3.11 with NumPy, SciPy, and pandas. Random seeds are fixed (`np.random.seed(42)`) throughout. The complete pipeline runs in under five minutes on commodity hardware; canonical values are verified against the published numbers by an automated test suite that runs on every repository commit.

3 Results

3.1 Non-random sequential structure

The full class transition matrix (Figure 1) departs significantly from independence: $\chi^2 = 1407.8$ with p effectively zero under a permutation null. The diagonal shows clear self-clustering on OT (2.05×), OK (1.57×), and QOK (1.56×). Off-diagonal, the CHEDY→QOK attractor (2.62×) and the AIIN→QOK repulsor (0.50×) are the most extreme cells.

3.2 Two distributed transition rules

Table 1 lists the two primary transition rules along with the observed count, expected count under independence, and split-half stability range.

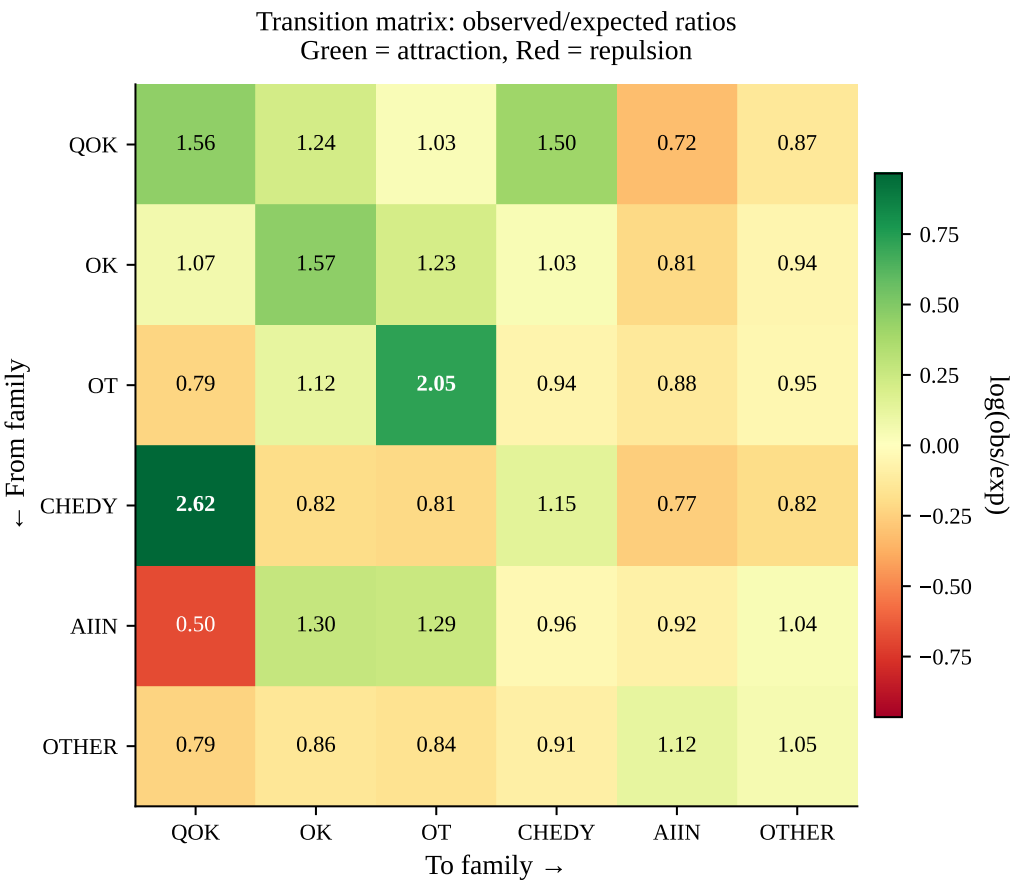

Figure 1: Transition matrix of observed/expected ratios between token families. Rows: source family; columns: destination family. CHEDY→QOK (2.62×) and AIIN→QOK (0.50×) are the two strongest off-diagonal departures from independence; the diagonal shows self-clustering on OT, OK, and QOK.

The rules are *distributed*, not phrasal. 77% of CHEDY tokens (with ≥ 5 occurrences) attract QOK, and the rule spans 369 unique CHEDY→QOK token pairs, with the top five pairs covering only 13.3% of the total. This is the signature of a class-level constraint rather than a few repeated collocations. Token shuffling within families preserves the effect perfectly ($\Delta = 0.00$), confirming that the rules describe class membership rather than specific token identity.

CHEDY→QOK is specific: after CHEDY, QOK is attracted at 2.62× but OK is mildly repelled (0.83×) and OT is mildly repelled (0.80×). The rule extends beyond immediate adjacency with decreasing strength (+1: 2.62×, +2: 1.23×, +3: 1.32×, +4: 1.16×, +5: 1.02×), reaching three to four tokens before decaying to baseline.

3.3 AIIN density invariance

Figure 2 summarizes AIIN density across sections and scribal hands. Per-section density varies substantially, from 4.1% (Astronomical) to 13.3% (Herbal A) and higher in Recipes Q20. This section-level variance would, in isolation, make AIIN look topic-dependent.

When pages are aggregated by Currier hand, the picture reverses. Currier A pages show mean AIIN density of 15.0% ($n = 102$); Currier B pages show 15.0% ($n = 72$). A two-sample Kolmogorov–Smirnov test on per-page distributions finds no significant difference (statistic $= 0.100$, $p = 0.742$). The bootstrap 95% confidence interval for the A−B difference is $[-1.95\%, +2.02\%]$.

AIIN is the only family with this property. All other backbone families (QOK, OK, OT, CHEDY) differ significantly between Currier A and B (all $p < 0.005$). The invariance is specifically a property of AIIN.

Table 1: Primary transition rules: observed/expected ratios with split-half stability ranges. Both are stable under every tested control (section splits, Currier A/B, line position, frequency conditioning, and random split-halves).

Rule	Ratio	Observed	Expected	Split-half range
CHEDY→QOK	2.625×	626	238.5	$[2.34, 2.67]$
AIIN→QOK	0.504×	160	317.4	$[0.39, 0.53]$

Function-word-consistent behavior. AIIN additionally shows two features typical of function words: it does *not* self-cluster (SC $\approx 0.98\times$), and it exhibits selective carry-through (Table 2) — passing content-family neighborhoods across an intervening AIIN for OK, OT, and CHEDY, but *blocking* QOK.

Table 2: Carry-through ratios: $X \to$ AIIN $\to X$ normalized by the base rate of X. AIIN passes OK, OT, and CHEDY neighborhoods through but blocks QOK.

Family	Carry-through	Interpretation
OK	2.73×	Strong carry-through
OT	2.21×	Strong carry-through
CHEDY	1.64×	Moderate carry-through
QOK	0.83×	Blocked

Caveat. Invariance holds under the standard AIIN definition (contains `aiin` or `ain`) but fails under a stricter definition (exact `aiin`, $p = 0.004$) and a looser definition (any `ain`/`in` stem, $p = 0.001$). The behavior is definition-sensitive, and we report only the standard-definition result as robust.

3.4 Bidirectional self-clustering symmetry

The most distinctive cross-linguistic result is summarized in Figure 3 and Table 3. Voynich clusters at $1.52\times$ by prefix and $1.54\times$ by suffix (ratio 0.99; bootstrap 95% CIs: prefix $[1.41, 1.62]$, suffix $[1.48, 1.60]$). It is the only tested system in the SYMM-HIGH bucket.

Earlier comparative analyses in this project that used only prefix-family self-clustering reported Arabic as the closest structural match. This finding is retracted: under bidirectional analysis, Arabic is strongly suffix-dominant ($P/S = 0.72$) and Voynich is symmetric ($P/S = 0.99$). The two systems occupy different structural categories. Similarly, earlier claims in this project that Uralic languages (Finnish, Estonian) match Voynich were based on 10K-sentence corpora that produced inflated prefix SC values; at 100K sentences, both Finnish and Estonian become SUFFIX-DOM. Ottoman Turkish, tested separately using a small historical UD treebank (16,890 words), falls in the SYMM-LOW region and is also not a match. Swahili (Bantu), which has textbook bidirectional morphology (prefix subject/class agreement and suffix tense/aspect markers), also tests SYMM-LOW (prefix $0.79\times$, suffix $0.96\times$, ratio 0.82). Georgian (Kartvelian), with polypersonal prefix+suffix agreement, tested SYMM-LOW (prefix $0.98\times$, suffix $0.86\times$). Tagalog (Austronesian), with infixation plus prefix/suffix morphology, also tested SYMM-LOW (prefix $0.44\times$, suffix $0.41\times$). Across four typologically distinct bidirectional-morphology languages (Swahili, Ottoman Turkish, Georgian, Tagalog), none approaches SYMM-HIGH. Mandarin Chinese (Sinitic), with isolating morphology and no affixes, tested SYMM-LOW (prefix $0.82\times$, suffix $0.78\times$, ratio 1.04), confirming that the SYMM-LOW region spans the full morphological spectrum from isolating to polysynthetic. Having bidirectional morphology in a natural language does not suffice to produce Voynich's symmetric elevated clustering.

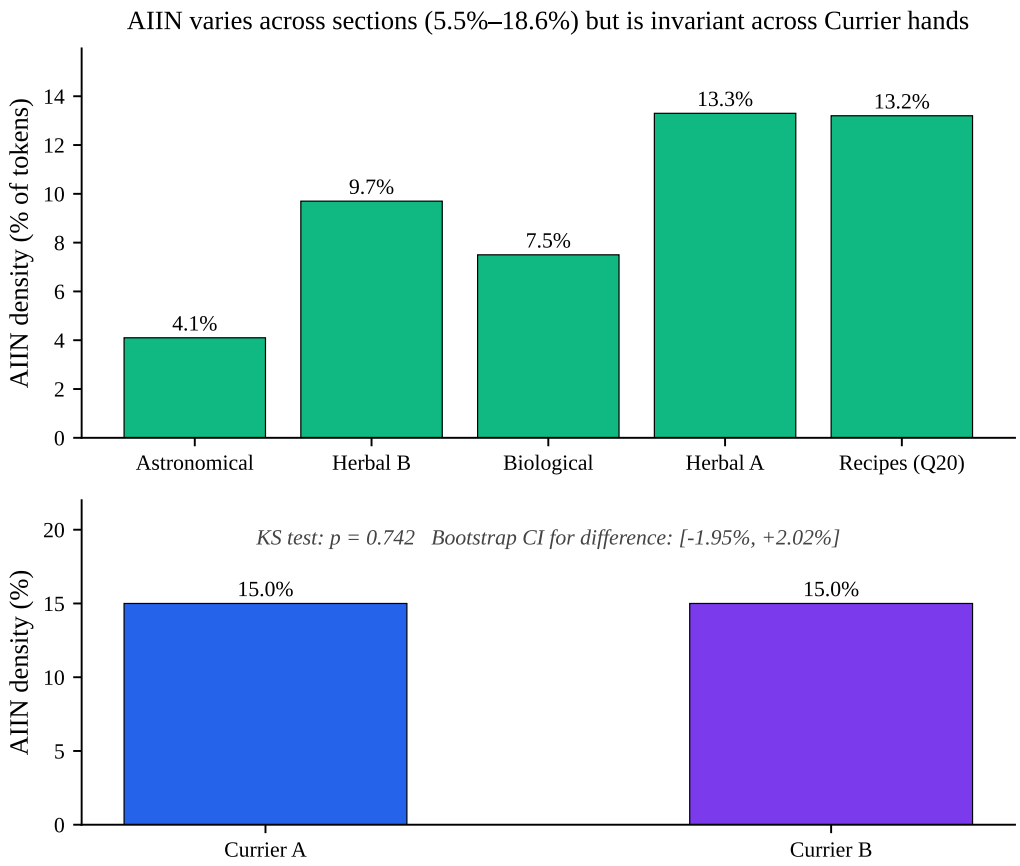

Figure 2: AIIN density by section (top) varies from 4.1% to 13.3%. The same families aggregated by Currier hand (bottom) are statistically indistinguishable: A $= 15.0\%$, B $= 15.0\%$, KS $p = 0.742$, bootstrap 95% CI for difference $[-1.95\%, +2.02\%]$.

3.5 Self-clustering is method-sensitive

The pooled-backbone self-clustering value is $1.451\times$ (CI $[1.34, 1.52]$). Aggregated over all six classes including OTHER, it is $1.384\times$. Page-level mean is $0.929\times$ — below unity. This range (0.93 to 1.45) reflects real methodological sensitivity: pooled computation lets a small number of high-density pages dominate, while page-level averaging treats each page equally but loses statistical power on short pages.

We report the pooled-backbone value as canonical for cross-linguistic comparison (the method applied identically to all systems) but emphasize that the magnitude is method-dependent. The *ratio* between prefix and suffix SC — the basis of the central result in §3.4 — is stable across all three methods.

3.6 Cross-transcription stability

A common critique of Voynich statistical work is that findings may be artifacts of a particular transliteration's tokenization decisions. To test this, we parsed the Landini–Stolfi Interlinear (LSI) file, which contains independent transcriptions by Currier, the First Study Group (Friedman), Takahashi, and Grove, all mapped to EVA alphabet but with each transcriber's own word-boundary and line-break judgments.

Table 4 shows that all four transcribers independently produce SYMM-HIGH bidirectional symmetry, CHEDY→QOK attraction in the range 2.01–2.48×, AIIN→QOK repulsion in the range 0.21–0.47×, and line-bounded grammar reset (within-line ratios exceed cross-line ratios for every transcriber). The structural findings are not artifacts of any particular transcriber's tokenization decisions. This result substantially closes the "tokenization artifact" attack vector: independent scholars working decades apart, with different approaches to ambiguous word boundaries, all recover the same structural signature.

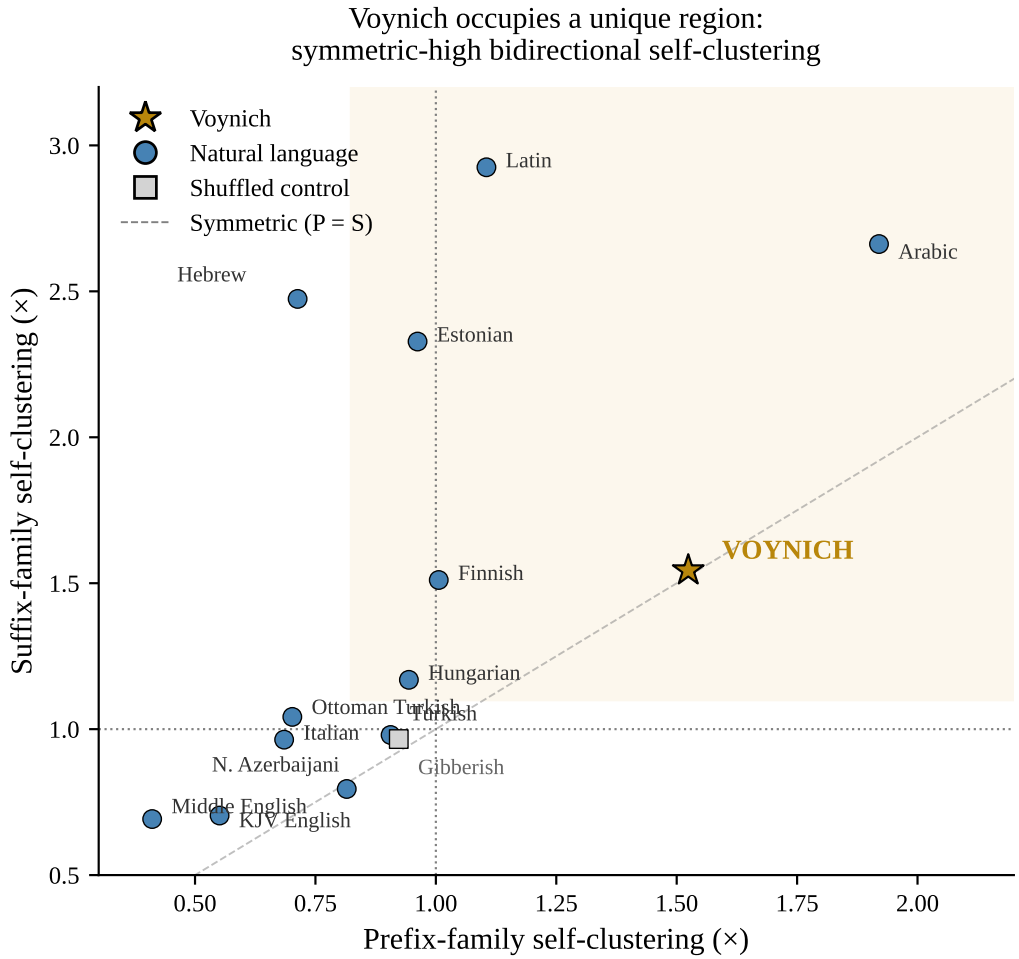

Figure 3: Prefix vs. suffix self-clustering across 18 tested systems. All natural languages with positive clustering (Arabic, Latin, Estonian, Hebrew, Finnish, Hungarian) are suffix-dominant. Voynich is the only system in the symmetric-high region (shaded). The shuffled control (`Gibberish`) sits near the origin.

3.7 Line-bounded grammar

The transition rules of §3.2 are strictly line-internal. Within-line CHEDY→QOK is 2.54×; across line boundaries it drops to 0.85×. Within-line AIIN→QOK is 0.40×; across lines it is 1.22×. Both rules reverse at line breaks.

The reset is universal: it occurs at every line-break type regardless of line length, section, or Currier hand, and is not specific to paragraph boundaries. In contrast, self-clustering *does* persist across lines (OK at 2.24×, OT at 3.03×, CHEDY at 1.50×). Family neighborhoods span line breaks; sequential grammar does not.

Whole-line family-composition templates recur at only 1.04× over a shuffled control (and 1.20× when measured by coverage rather than exact match), substantially lowering the plausibility of a simple template-library model. Lines have unique family compositions, constrained by within-line grammar but not drawn from a fixed pool. This pattern is consistent with lines functioning as clause-like or phrase-like grammatical units, though we do not claim a literal clause identification.

3.8 Suffix-led multi-feature agreement

Adjacent tokens within specific family pairings show above-chance agreement on word-final character class (suffix agreement ratios 1.18–1.75, Table 5). The effect compounds when multiple features are jointly tracked: for OK→OT pairs, suffix agreement alone is 1.75×, suffix + length is 3.44×, and suffix

Table 3: Prefix/suffix self-clustering across tested systems. Voynich is the only SYMM-HIGH system; all other natural languages with positive clustering are SUFFIX-DOM. Ottoman Turkish (UD treebank, 16,890 words) Swahili, Georgian, and Tagalog (all Leipzig 100K) also tested SYMM-LOW despite having bidirectional morphology. Mandarin (Sinitic, isolating) also tested SYMM-LOW, establishing the morphologically minimal floor.

System	Prefix SC	Suffix SC	P/S	Bucket
Voynich	**1.524**	**1.544**	**0.99**	**SYMM-HIGH**
Arabic	1.920	2.662	0.72	SUFFIX-DOM
Latin	1.105	2.925	0.38	SUFFIX-DOM
Estonian	0.962	2.328	0.41	SUFFIX-DOM
Hebrew	0.713	2.474	0.29	SUFFIX-DOM
Finnish	1.006	1.511	0.67	SUFFIX-DOM
Hungarian	0.944	1.169	0.81	SUFFIX-DOM
Ottoman Turkish	0.702	1.042	0.67	SYMM-LOW
Swahili	0.791	0.960	0.82	SYMM-LOW
Georgian	0.978	0.856	1.14	SYMM-LOW
Tagalog	0.444	0.411	1.08	SYMM-LOW
Mandarin (Pinyin)	0.817	0.784	1.04	SYMM-LOW
Turkish	0.906	0.980	0.92	SYMM-LOW
Italian	0.685	0.964	0.71	SYMM-LOW
N. Azerbaijani	0.815	0.795	1.02	SYMM-LOW
KJV English	0.551	0.704	0.78	SYMM-LOW
Middle English	0.411	0.692	0.59	SYMM-LOW
Gibberish (control)	0.923	0.966	0.96	SYMM-LOW

+ length + mantle + circles combined is 8.74×.

Length agreement conditioned on suffix and frequency collapses to 1.00×: length similarity is partially a downstream consequence of suffix similarity. Suffix is the primary agreement feature.

CHEDY selects specific QOK subtypes ($\chi^2 = 36.4$, $p = 7.2 \times 10^{-5}$): `-dy` CHEDY tokens attract 40% `-dy` QOK tokens; `-ey` CHEDY attracts 32% `-ey` QOK. This is concord of a specific, non-trivial kind: the attractor and the attracted must share suffix form.

3.9 Agreement cascades through three-token chains

When two adjacent tokens (A, B) agree on suffix, agreement for the next position $(B \to C)$ jumps dramatically. For a CHEDY→OTHER→CHEDY chain, the conditional probability of $B \to C$ agreement is 85% if $A \equiv B$ in suffix, versus 4% if $A \not\equiv B$ — a +81 percentage-point cascade. Similar conditional lifts hold for other triples (QOK→OTHER→QOK: +44pp; QOK→QOK→QOK: +33pp; CHEDY→QOK→CHEDY: +56pp).

Suffix agreement also jumps over intervening `OTHER`-family tokens: QOK→[OTHER]→QOK at 2.32× base rate; CHEDY→[OTHER]→CHEDY at 1.43×; OT→[OTHER]→OT at 1.87×. The agreement system treats OTHER-family tokens as transparent for the purposes of morphological concord. This is the quantitative signature of long-distance agreement.

3.10 Productive morphological paradigms

Each family contains an edit-distance graph with identifiable high-frequency hubs. Of 50 tested types per family, all are connected with no isolated nodes. Top hubs: `chedy` connects to 12 edit-1 neighbors, `daiin` to 10, `qokeedy` to 7.

Four edit operations recur across all major sections and are positionally locked:

Table 4: Cross-transcription stability. All four independent transcribers produce SYMM-HIGH bidirectional symmetry and CHEDY→QOK attraction well above chance. Line-bounded grammar (within-line > cross-line) is universal.

Transcriber	Tokens	C→Q	A→Q	Pfx SC	Sfx SC	Bucket
Currier	16,453	2.48×	0.42×	1.30×	1.53×	SYMM-HIGH
FSG (Friedman)	28,811	2.40×	0.45×	1.23×	1.47×	SYMM-HIGH
Takahashi	30,426	2.41×	0.47×	1.28×	1.50×	SYMM-HIGH
Grove	7,657	2.01×	0.21×	1.14×	1.15×	SYMM-HIGH
ZL (baseline)	31,608	2.63×	0.50×	1.52×	1.54×	SYMM-HIGH

Table 5: Single-feature suffix agreement (observed same-ending divided by expected under independence). All ratios significant vs. a shuffled null ($z = 3.4$–5.0).

Family pair	Suffix agreement
OK→OT	1.75×
OK→OK	1.67×
OT→OT	1.49×
QOK→QOK	1.41×
CHEDY→QOK	1.18×

- Insertion/deletion of `e`, 92% mid-position;
- Insertion/deletion of `d`, 74% end-position;
- Substitution `c`↔`s` (as in `ch`/`sh`), 62% mid-position;
- Insertion/deletion of `l`, 89% start-position.

The same operations apply to distinct vocabulary in distinct sections, producing section-appropriate variants while preserving paradigm structure.

High-frequency stems generate more edit-1 variants than low-frequency stems. The log-frequency vs. variant-count correlation is $r = 0.52$ (QOK), $r = 0.60$ (CHEDY), $r = 0.69$ (AIIN). Stems at frequency ≥ 100 have 2–3× more variants than stems at frequency 2–5. This frequency-productivity correlation is a standard property of natural-language morphology (productive paradigms generate more variants for more-common lemmas) and is, to our knowledge, not a property of tested constructed or cipher systems.

3.11 Glyph-layer architecture

Decomposing tokens into three glyph layers — the *crust* (first and last characters), the *mantle* (alternating middle characters), and the *core* (gallows and inner structure) — reveals that different layers carry different information:

- **Clustering** lives independently in both crust-only ($1.86\times/2.01\times$) and mantle+core-only ($1.88\times/2.00\times$) layers. Full-token values ($1.29\times/1.54\times$) are diluted by cross-layer interference.
- **Sequential grammar** lives primarily in circles (o, a, y) and core (gallows). Scrambling circles drops CHEDY→QOK from 2.50× to 1.75×. Scrambling crust has no effect on transitions; scrambling mantle has no effect.
- **Family identity** is best predicted by crust characters (79.4% accuracy). QOK and OK are 100% predicted by their first character. CHEDY and AIIN are 99% predicted by their last character.

Different glyph zones carry structurally different information: *identity* in the crust, *grammar* in the circles and core. This is the quantitative form of an architectural claim about what the glyphs are doing.

4 Discussion

4.1 What the findings establish

The combination of Sections 3.2–3.4 substantially lowers the plausibility of three classes of explanation for the Voynich text, relative to the alternative that the text encodes a structured language system.

Randomness is substantially less plausible. The χ^2 statistic of the class transition matrix is $\sim 40\times$ that of a shuffled control, and distributed agreement cascades of the form documented in §3.3–§3.9 are not produced by random sequences.

Simple table-generation schemes are substantially less plausible. A token stream drawn from a fixed family-composition template library would exhibit whole-line template recurrence well above chance. The observed value is $1.04\times$. Moreover, such a scheme cannot generate the observed productive morphology-like paradigm structure (§3.10).

Simple substitution ciphers (monoalphabetic, polyalphabetic over a Latin base) are substantially less plausible. These preserve source-language suffix dominance and cannot produce bidirectional symmetry. Voynich's $P/S = 0.99$ is structurally distinct from any tested natural language's profile. This does not exclude more elaborate cipher schemes involving, for example, syllabic encoding or bidirectional padding, which have not been exhaustively tested here.

4.2 What the findings do *not* establish

These results do not identify Voynich as any particular natural language, because no tested natural language matches Voynich's symmetric-high profile. They do not identify it as artificial either: the space of possible natural languages is larger than our 16-language sample, and a language with bidirectional morphology (for example, a polysynthetic language or a combinatorial morphology with comparable prefix and suffix inventories) could in principle produce this signature. The findings constrain the space of viable explanations; they do not localize a single one.

The bidirectional symmetry also does not directly imply meaning. It is a property of the token sequences, compatible with multiple semantic interpretations.

4.3 Comparison to previous statistical analyses

Previous statistical work on Voynich (surveyed in, e.g., Bowern and Lindemann, 2021; Reddy and Knight, 2011) has repeatedly confirmed that the text is non-random and has word-like Zipfian distribution. The contribution of the current work is (i) the bidirectional self-clustering framework that removes a measurement bias present in prefix-only analyses, and (ii) the checklist of structural constraints that any proposed explanation must simultaneously satisfy (§5). We treat the statistical results not as a solution but as a scaffolding against which future proposed solutions can be evaluated.

5 Minimum Viable Explanation: A Checklist

Any proposed explanation of the Voynich text — decipherment, generation theory, cipher identification, or hoax mechanism — must simultaneously account for *all* of the following. Explanations satisfying some but not others are incomplete.

1. **Line-bounded grammatical units.** Within-line CHEDY→QOK is $2.54\times$; across lines it drops to $0.85\times$. Transition rules are line-internal and reset at every line boundary, consistent with lines functioning as clause-like or phrase-like grammatical units.

2. **CHEDY→QOK class specificity.** CHEDY attracts QOK at 2.62× while simultaneously repelling OK (0.83×) and OT (0.80×). This is targeted, not a general content-density effect.
3. **Suffix-led multi-feature agreement.** Adjacent tokens agree on suffix at 1.18–1.75×, compounding to 5–9× when suffix, length, and mantle agreement are jointly tracked.
4. **Agreement cascades through three-token chains.** Conditional lifts of 33–80 percentage points, including over intervening OTHER-family tokens.
5. **Productive morphological paradigms.** Hub-centered edit-distance graphs, the same four edit operations recurring across all sections, and a log-frequency vs. variant-count correlation of $r = 0.52$–0.69.
6. **Bidirectional self-clustering symmetry.** $P/S = 0.99$ with both prefix and suffix SC $> 1.5\times$. Unique among 18 tested systems.
7. **Section-stable grammar with shifting lexicon.** Transition rules, agreement, paradigm operations, and family proportions are consistent across sections, but Jaccard overlap of family vocabulary between sections is 0.09–0.25.
8. **Open vocabulary, natural distribution.** 71.4% hapax legomena; type/token ratio 0.23; grammatical variance coefficient 0.40; top-100 bigrams cover only 3.7% of text.

Table 6 evaluates four candidate explanation classes against these requirements.

Table 6: Minimum viable explanation checklist, applied to four candidate classes. "Y" means compatible with the requirement; "N" means not compatible; "?" means possible given significant design effort without a known 15th-century precedent. Encoded structured language (plausibly natural language) is the strongest current candidate class in this comparison framework.

Requirement	Encoded NL	Constructed	Simple cipher	Table/grille
1. Line-bounded grammatical units	Y	?	N	N
2. Class specificity (CHEDY→QOK)	Y	?	N	N
3. Suffix-led multi-feature agreement	Y	?	N	N
4. Three-token agreement cascades	Y	?	N	N
5. Productive morphology-like paradigms	Y	N	N	N
6. Bidirectional symmetry	?	?	N	N
7. Stable grammar, shifting lexicon	Y	?	Y	Partial
8. Open, natural-distribution vocabulary	Y	?	Partial	N

Of the candidate classes tested, encoded structured language — plausibly natural language — is the only one compatible with all eight requirements without invoking mechanisms that lack known 15th-century precedent. This does not prove encoded natural language, nor does it exhaust the space of possibilities. The candidate set is limited to four broad classes and the comparator pool to 16 natural languages across 9 families; other natural-language types (Sinitic, polysynthetic, historical shorthand) and more elaborate constructed systems remain untested. The checklist is offered as a diagnostic framework, not as decisive proof: it defines what any successful explanation must account for and substantially raises the bar on randomness, simple table-based generation, and simple substitution ciphers. The strongest current explanation consistent with the full checklist is encoded structured language.

6 Limitations

Nine limitations apply to all of the above and should not be minimized.

1. EVA-transcription dependence. Family definitions (QOK, OK, OT, CHEDY, AIIN) are based on the EVA alphabet. They may not correspond to meaningful character-level units in the original script. Results under FSG or Currier transcription alphabets have not been tested.

2. Modern comparison corpora. The nine Leipzig corpora are modern Wikipedia text. Structural proxies for their language families, not medieval equivalents. Historical Latin, Old French, and medieval Italian are not tested. Ottoman Turkish was tested with a small UD corpus (SYMM-LOW) but a larger historical corpus is needed.

3. No decipherment. This analysis identifies statistical patterns in token sequences. It does not decode any text, identify any language, or assign meaning to any word.

4. Method sensitivity. Self-clustering values range from 0.93× (page-level) to 1.45× (pooled backbone) depending on computation method. The prefix/suffix ratio is stable but uses auto-detected affix families that may vary across runs.

5. Directional bias. The prefix-family method used in earlier Voynich research inherently favors languages with consistent prefixing morphology. The suffix extension introduced here corrects this bias but introduces its own auto-detection uncertainty.

6. AIIN invariance is definition-dependent. Holds under the standard definition but fails under stricter or looser alternatives.

7. Small Ottoman Turkish corpus. The UD treebank's 16,890 words show SYMM-LOW but this result should be confirmed on a larger historical corpus before being taken as definitive. Swahili (Leipzig 100K) also tested SYMM-LOW despite bidirectional morphology.

8. Cross-transcription validation. Results are stable across four independent EVA-based transcribers (Currier, FSG, Takahashi, Grove; see §3.6). Validation under non-EVA alphabets (original Currier alphabet, original FSG alphabet with different character boundaries) remains an open question.

9. Reproducibility is partial. Three of the core analyses (transition rules, AIIN invariance, bidirectional self-clustering) are covered by reproducible scripts with automated regression tests (`run_all.py`, `tests/test_canonical_values.py`). Several of the more detailed analyses reported here — line-bounded grammar, suffix-agreement cascades, paradigm-graph morphology, and the glyph-layer ablations — were developed interactively during the research session and are documented in the project's durable findings framework (`docs/durable_findings.md`) with their statistical methods and canonical values, but are not yet packaged as standalone scripts. Scripting these analyses is planned work prior to any peer-reviewed journal submission.

7 Open Problems

Three concrete extensions would substantially strengthen or refute these findings:

1. **Non-EVA alphabet validation.** Cross-transcription stability has been confirmed across four EVA-based transcribers (§3.6). The remaining open question is whether findings hold under the original Currier and FSG character-boundary definitions, which assign different glyph groupings than EVA.

2. **Ottoman Turkish with larger corpus.** The UD treebank shows SYMM-LOW but is too small to be definitive. Swahili (Bantu) also tested SYMM-LOW despite rich bidirectional morphology. A historical corpus of 15th-century Ottoman Turkish remains worth testing but is no longer the sole candidate.

3. **Additional language types.** Sinitic has been tested (Mandarin, SYMM-LOW). Polysynthetic languages (Nahuatl — unavailable at corpus scale from Leipzig) and historical shorthand systems (Tironian notae) remain untested. Georgian, Tagalog, and Mandarin have been tested and all fall in SYMM-LOW.

8 Data and Code Availability

All analysis code, frozen datasets ($\sim$ 228 MB with SHA-256 checksums), and result files are available at `https://github.com/amy2213/Voynich-Transition-Grammar` under MIT license for code and original data licenses for corpora (Voynich: public domain; Leipzig: CC-BY; Gutenberg: public domain).

The pipeline runs via `python run_all.py` and covers the three core analyses reported in §3 subsections 3.1 through 3.4. An automated test suite (`pytest tests/`) verifies that the canonical values for these analyses match the output of the current pipeline within stated tolerances, and a continuous integration workflow runs the validation and test suite on every commit. Analyses reported in §3 subsections 3.6 through 3.10 — line-bounded grammar, suffix-agreement cascades, paradigm morphology, and glyph-layer architecture — are documented in `docs/durable_findings.md` with their statistical methods, but are not yet packaged as standalone scripts. Scripting these analyses is planned work.

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